

Fundamentals of Electric Circuits

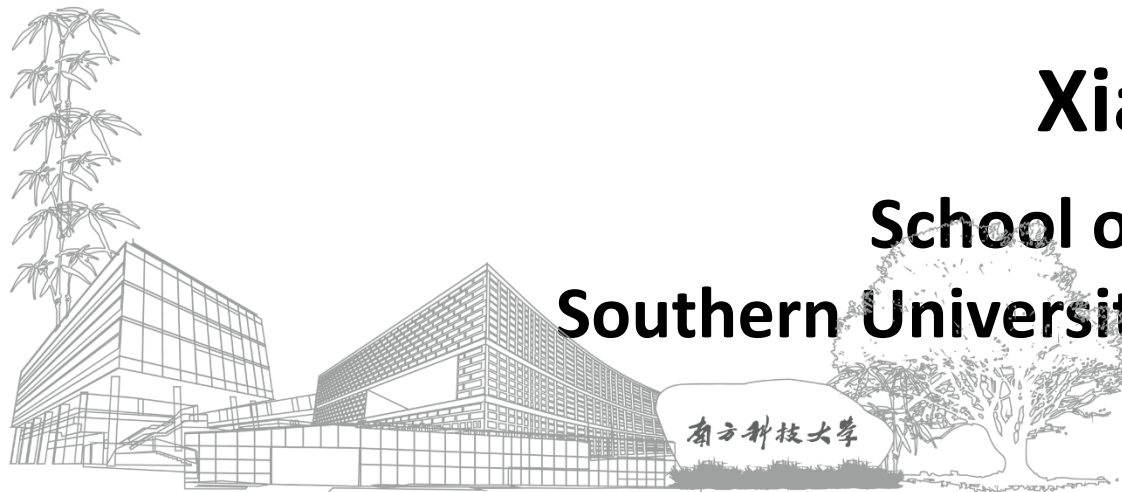
电路基础

Lecture4: Circuit Theorems

Xiaolong Liu

School of Microelectronics

Southern University of Science and Technology

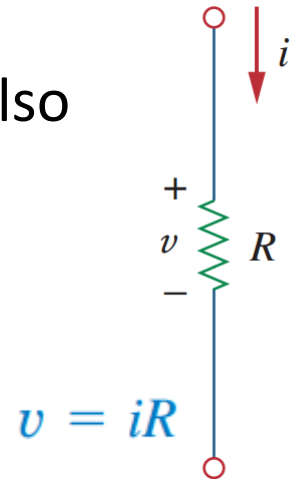
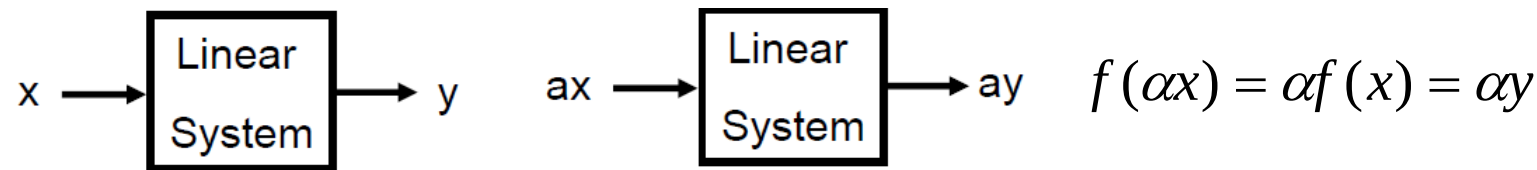


Outline

- **Linearity**
- **Superposition**
- **Source Transformation**
- **Thevenin's Theorem**
- **Norton's Theorem**
- **Maximum Power Transfer**

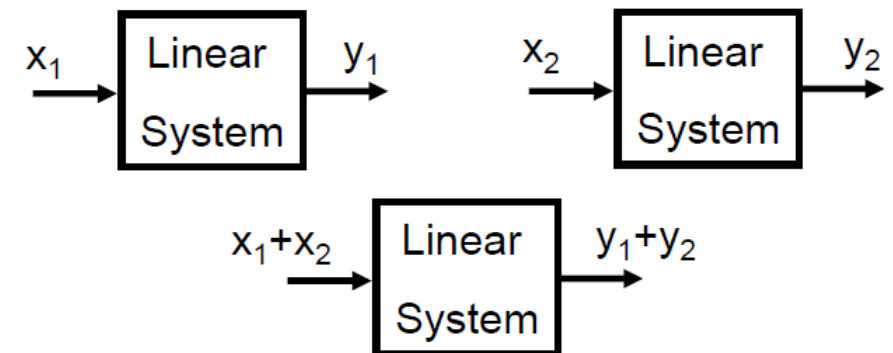
Linearity (I)

- A **linear circuit** (线性电路) is a circuit whose output is linearly related (or directly proportional) to its input
- A system is linear if and only if it has these two properties
 - If the input of the system is multiplied by a constant, then the output is also multiplied by a same constant (**Homogeneity**, 齐次性)



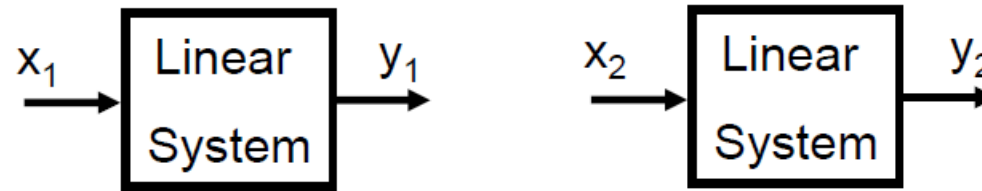
- The output of the system to a sum of inputs is the sum of the outputs to each input applied separately (**Additivity**, 可加性)

Let $\begin{cases} y_1 = f(x_1) \\ y_2 = f(x_2) \end{cases}$, We have $f(x_1 + x_2) = y_1 + y_2$

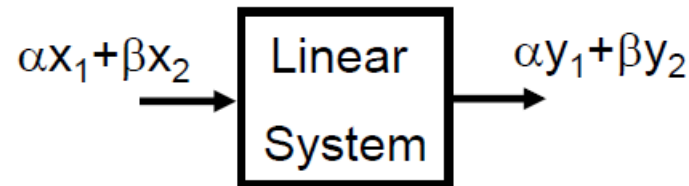


Linearity (II)

- A linear circuit consists of only **linear elements**, **linear dependent sources**, and **excited by independent sources**
- We consider only linear circuits in this course
- General definition



- For all α and β



$$f(\alpha x_1 + \beta x_2) = \alpha y_1 + \beta y_2$$

Example: Linearity Property (I)

- For the circuit in Fig. 4.2, find I_o when $v_s = 12\text{ V}$ and $v_s = 24\text{ V}$

Solution:

Applying KVL to the two loops, we obtain

$$12i_1 - 4i_2 + v_s = 0$$

$$-4i_1 + 16i_2 - 3v_x - v_s = 0$$

Together with $v_x = 2i_1$, we get

$$i_2 = \frac{v_s}{76}$$

So, for $v_s = 12\text{ V}$, $I_o = i_2 = 3/19 = 0.158\text{ A}$;

for $v_s = 24\text{ V}$, $I_o = i_2 = 6/19\text{ A} = 0.316\text{ A}$

→ This shows that when the source value is doubled, the output I_o is doubled too

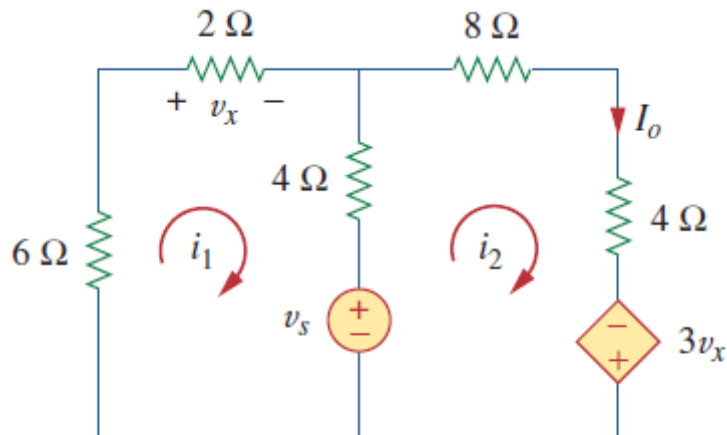


Figure 4.2
For Example 4.1.

Example: Linearity Property (II)

- Q: For the circuit in Fig. 4.3, find v_o when $i_s = 30$ A and $i_s = 45$ A.

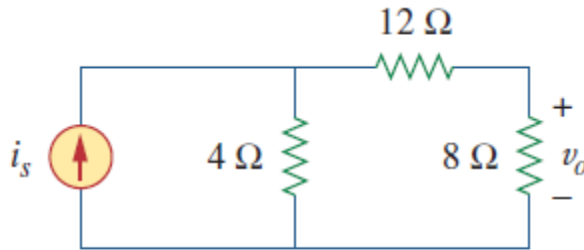


Figure 4.3
For Practice Prob. 4.1.

First assume $v_o = 8$ V, we get $i_s = 6$ A

According to the linearity, we obtain

$v_o = 40$ V when $i_s = 30$ A

$v_o = 60$ V when $i_s = 45$ A

Superposition (I)

- If there are **two or more independent sources** there are two ways to solve for the circuit parameters:
 - Nodal or mesh analysis
 - Use superposition
- The idea of superposition rests on the linearity property
- The **superposition principle (叠加定理)** states that the voltage across (or current through) an element in a linear circuit is the **algebraic sum (代数和)** of the voltages across (or currents through) that element due to **each independent source acting alone (每个独立源单独作用)**.
- **Using superposition means applying one independent source at a time.** Dependent sources are left alone since they are controlled by circuit variables

Superposition (II)

- Steps for applying superposition principle
 1. Turn off all independent sources except one source. Find the desired voltage or current due to that source
 2. Repeat the previous step for each of the independent sources
 3. Find the total desired voltage or current by algebraically adding the contributions due to each of the independent sources

Q: Use the superposition theorem to find v in the circuit of Fig. 4.6.

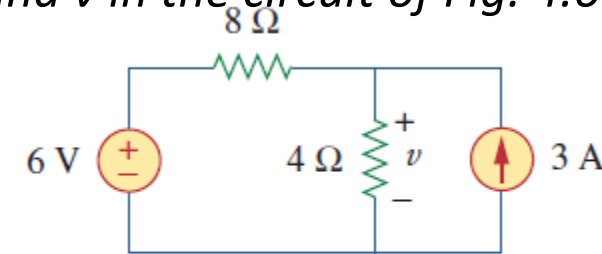


Figure 4.6
For Example 4.3.

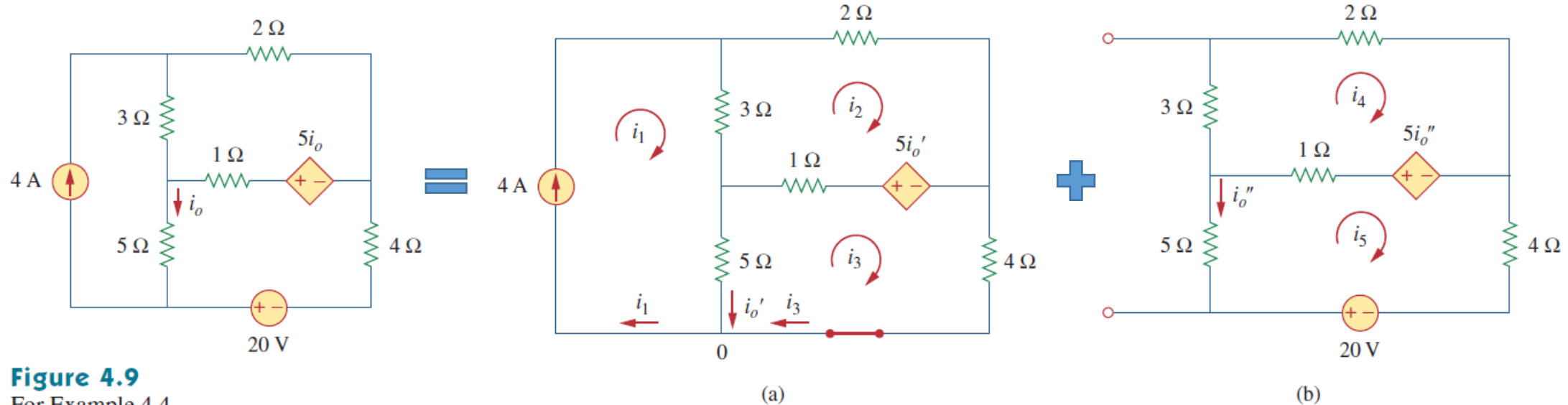
Solution:

- Since there are two sources, let $v = v_1 + v_2$ where v_1 and v_2 are the contributions due to the 6 V voltage source and the 3 A current source, respectively
- Turn off the 3 A current source, $v_1 = 6 \cdot 4 / (4 + 8) = 2$ V
- Turn off the 6 V voltage source, $v_2 = 3 \cdot (4 \parallel 8) = 8$ V
- The total contribution adds up to $v = v_1 + v_2 = 10$ V

Turn off a source means to shorten (短路) voltage sources (set to 0 V) and/or to open (开路) current sources (setting to 0 A)

Example: Superposition (I)

- Q: Find i_o in the circuit of Fig. 4.9 using superposition



Solution:

The circuit in Fig. 4.9 involves a dependent source, which must be left intact

let $i_o = i_o' + i_o''$, where i_o' and i_o'' are due to the 4 A current source and 20 V voltage source, respectively
To obtain i_o' we turn off the 20 V source so that we have the circuit in Fig. 4.9(a)

To obtain i_o'' we turn off the 4 A current source so that the circuit becomes that shown in Fig. 4.9(b)

By mesh analysis, we get $i_o' = 52/17$ A, and $i_o'' = -60/17$ A. So, $i_o = i_o' + i_o'' = -8/17 = -0.471$ A

Example: Superposition (II)

- Q: Use the superposition theorem to find i in Fig. 4.12

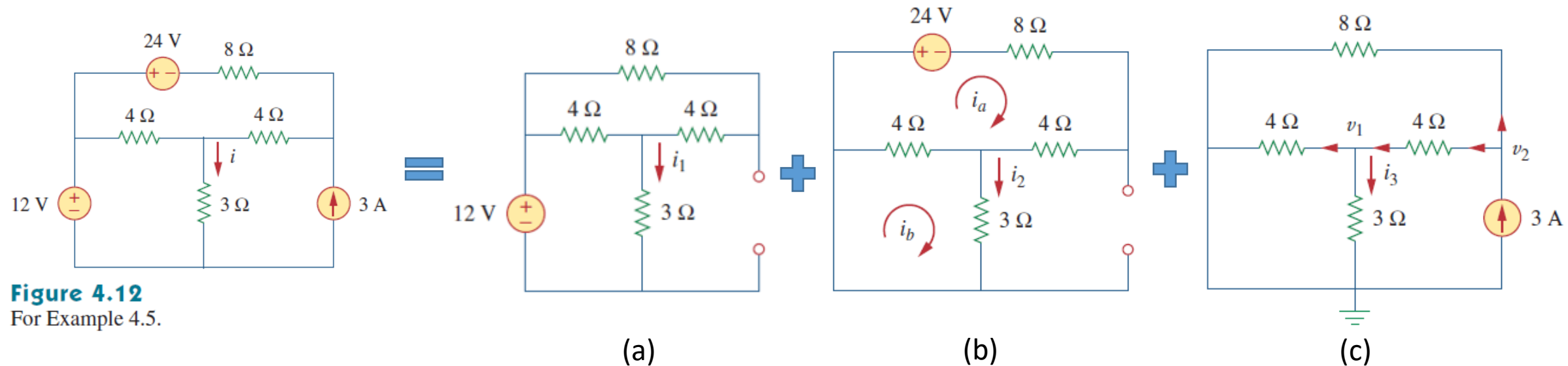


Figure 4.12
For Example 4.5.

Solution:

According to superposition theorem, Fig. 4.12 (a), (b) and (c)

We can find out $i_1 = 2 \text{ A}$, $i_2 = -1 \text{ A}$, and $i_3 = 1 \text{ A}$

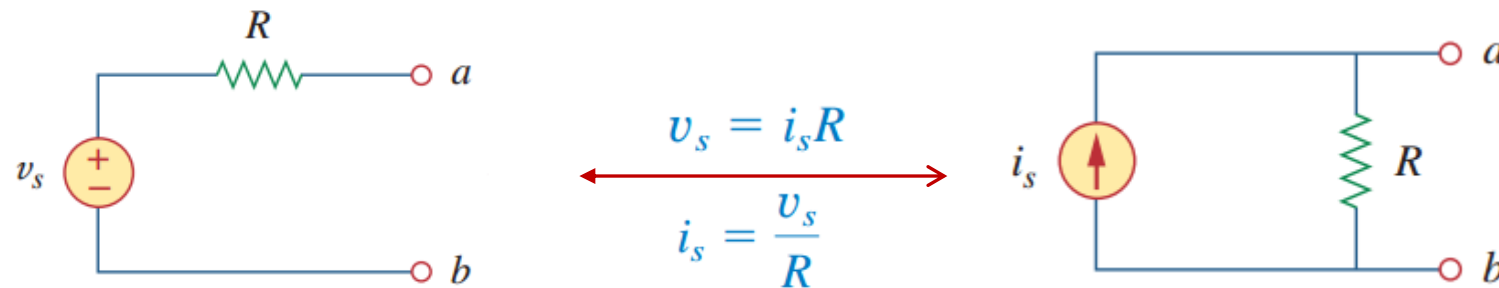
Therefore, $i = i_1 + i_2 + i_3 = 2 \text{ A}$

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- Linearity
- Superposition
- **Source Transformation**
- **Thevenin's Theorem**
- **Norton's Theorem**
- Maximum Power Transfer

Source Transformation

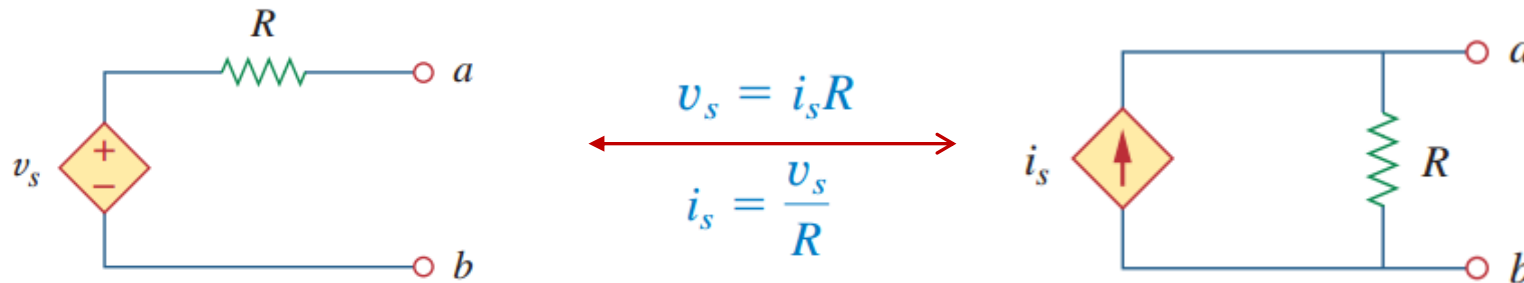
- Series-parallel combination and wye-delta transformation help simplify circuits
- **Source transformation (电源变换)** is another tool for simplifying circuits
- Source transformation refers to the process of replacing a **voltage source in series with a resistor R** with a **current source in parallel with a resistor** (with the same resistance value as R) and vice versa
 - The principle behind all of these transformations is **equivalence (等效)**. An equivalent circuit is one whose **v-i characteristics (伏安特性)** are **identical (相同)** with the original circuit



- If the sources are turned off the resistance at the terminals are both R .
- If the terminals are short circuited, the currents need to be the same, so $v_s = i_s R$, $i_s = v_s / R$

Source Transformation Theorem

- A source transformation does not affect the remaining part of the circuit
- When dealing with source transformation, we should keep the following points in mind
 - The arrow of the current source is directed towards the **positive terminal** of the voltage source
 - Source transformation is not possible when $R = 0$ for an ideal voltage source
 - For an ideal current source, $R = \infty$ also prevents the use of source transformation
- Source transformation also applies to dependent sources
 - But the dependent variable must be handled carefully



Example: Source Transformation (I)

- Use source transformation to find v_o in the circuit of Fig. 4.17

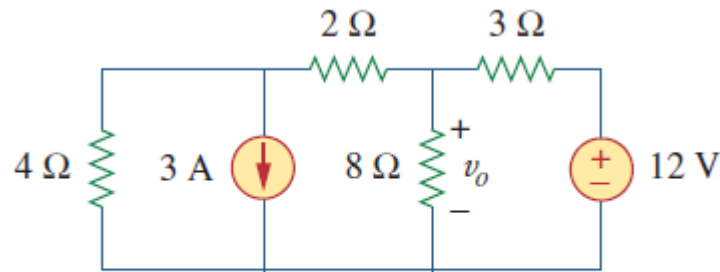


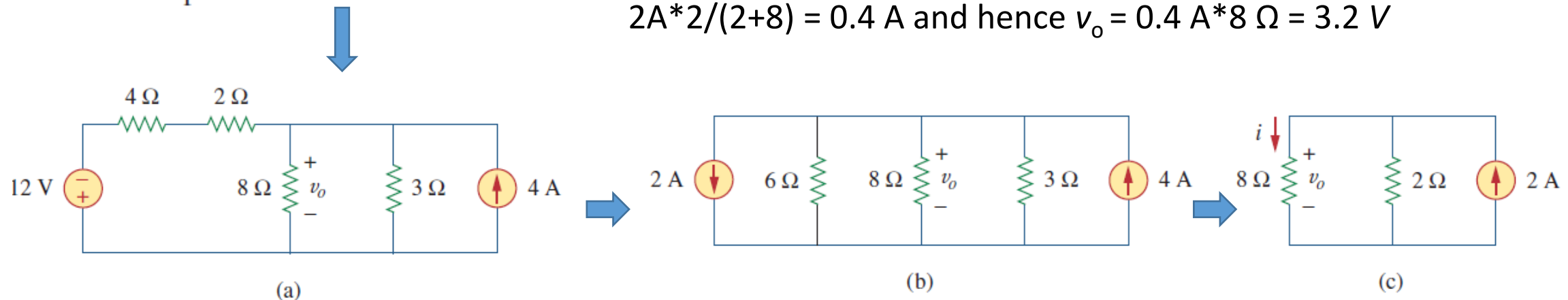
Figure 4.17
For Example 4.6.

Solution:

Transforming the $12\text{ V} + 3\ \Omega$ and $3\text{ A} \parallel 4\ \Omega$ sources, we get the equivalent circuit in Fig. (a)

Transforming the transformed $12\text{ V} + 6\ \Omega$ again, we get Fig. (b)

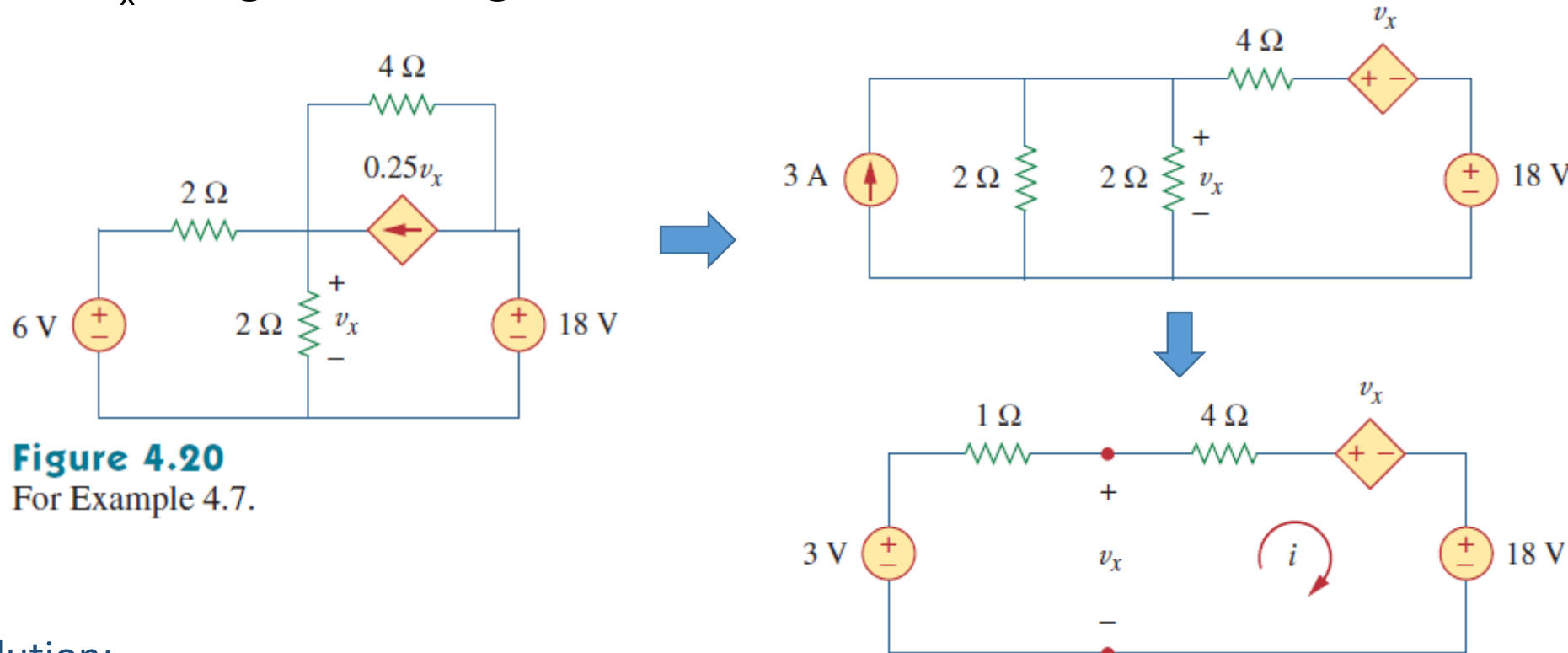
Simplifying it, we get Fig. (c). By current division we achieve $i = 2\text{ A} \cdot 2 / (2 + 8) = 0.4\text{ A}$ and hence $v_o = 0.4\text{ A} \cdot 8\ \Omega = 3.2\text{ V}$



Q: How would you solve the problem w/o source transformation?

Example: Source Transformation (II)

- Find v_x in Fig. 4.20 using source transformation



Solution:

Transforming the VCCS $|| 4\Omega$, $6\text{ V} + 2\Omega$ sources, we get Fig. 4.21 (a)

Transforming the $3\text{ A} || (2\Omega || 2\Omega)$ source, we get Fig. 4.21 (b)

Applying KVL to the loop, we get $-3 + 5i + v_x + 18 = 0$. Together with $v_x = 3 - 1 \cdot i$, we get $i = -4.5\text{ A}$ and $v_x = 7.5\text{ V}$

Thévenin's Theorem (I)

- In practice, it often happens that a particular element (or block) in a circuit is variable (this variable block is usually called load) and the rest of the circuit elements are fixed
- Thevenin's theorem (戴维南定理)** provides a very powerful technique by which the fixed part of the circuit is replaced by an equivalent circuit (等效电路替代不变部分)
 - In 1883, Leon Charles Thévenin (1857-1926), a French telegraph engineer came up with the interesting idea of replacing a complicated linear circuit by an equivalent circuit consisting of an independent voltage source in series with a resistor
- Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a **voltage source V_{Th}** in series with a **resistor R_{Th}**

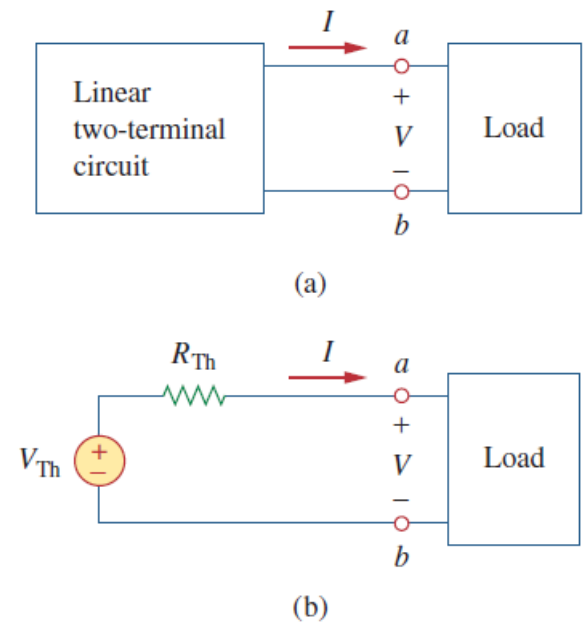
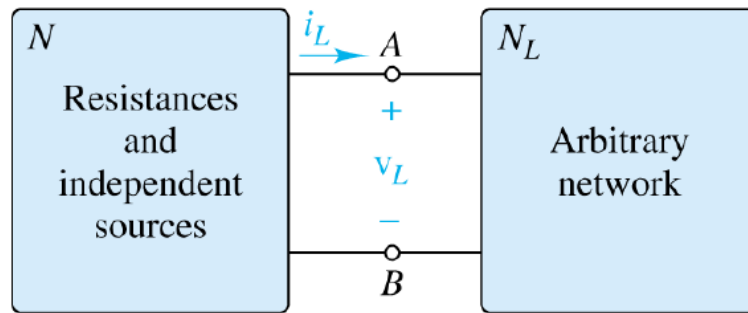


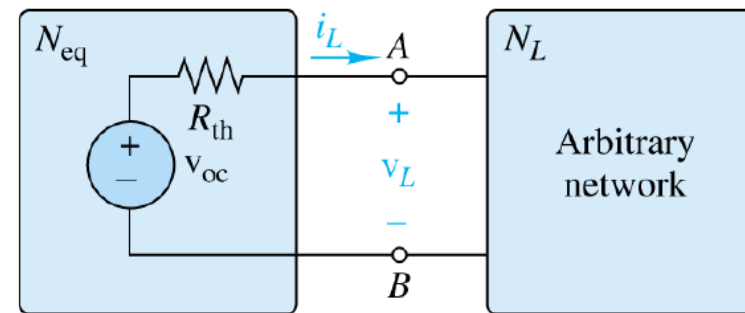
Figure 4.23
Replacing a linear two-terminal circuit by its Thevenin equivalent: (a) original circuit, (b) the Thevenin equivalent circuit.

Thévenin's Theorem (II)

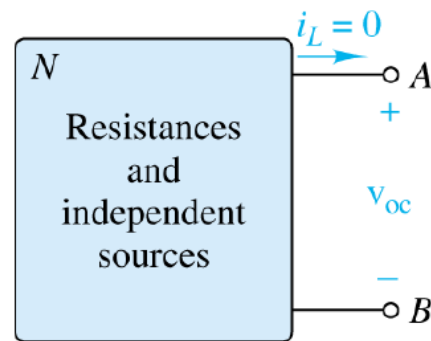
- The voltage source V_{Th} is the **open-circuit voltage** (开路电压) at the terminals and R_{Th} is the **input or equivalent resistance** (输入或等效电阻) measured at the terminals when **all the independent sources are turned off** (所有独立源关闭)



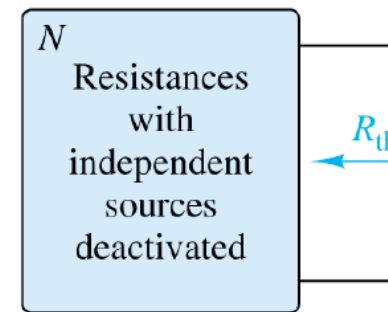
(a)



(b)



(c)



(d)

Thévenin's Theorem (III)

- There are two cases to consider when finding the equivalent resistance R_{Th}
 - **Case 1:** If there are **no dependent sources**, then the resistance may be found by simply turning off all the sources
 - **Case 2:** If there are **dependent sources**, we still turn off all the independent sources
 - ✓ Apply a voltage v_o (or current i_o) to the terminals and determine the current i_o (voltage v_o). Then $R_{Th} = v_o/i_o$
 - ✓ It is possible for the result of this analysis to end up with a negative resistance. This implies the circuit is supplying power, a reasonable case with dependent sources
- The Thevenin equivalent makes working with variable loads much easier

Example: Thévenin's Theorem (I)

- Find the Thevenin equivalent circuit of the circuit shown in Fig. 4.27, to the left of the terminals a-b. Then find the current through $R_L = 6, 16$ and 36Ω

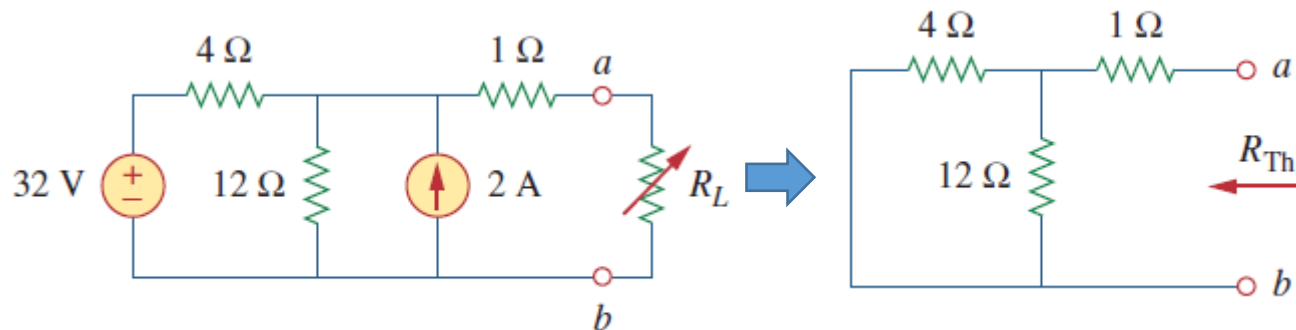
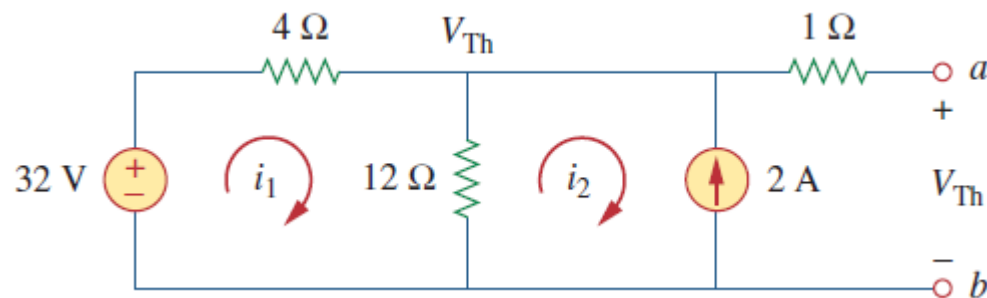


Figure 4.27
For Example 4.8.



Solution:

Using the circuits, V_{Th} and R_{Th} in the Thevenin equivalent circuit can be found to be:

$$R_{Th} = 1 + 4 \parallel 12 = 4 \Omega$$

$$V_{Th} = 32 \cdot 12 / (12 + 4) + 2 \cdot (4 \parallel 12) = 30 \text{ V}$$

Therefore, $i_L = 30 / (4 + R_L)$, which is 3, 1.5 and 0.75 A for $R_L = 6, 16$ and 36Ω , respectively.

Example: Thévenin's Theorem (II)

- Find the Thevenin equivalent of the circuit in Fig. 4.31 at terminals a-b

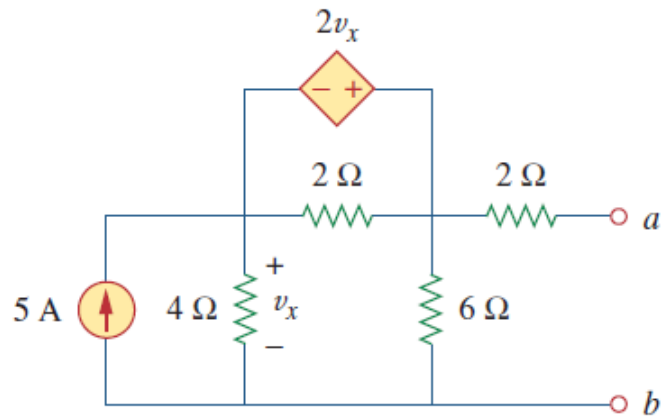
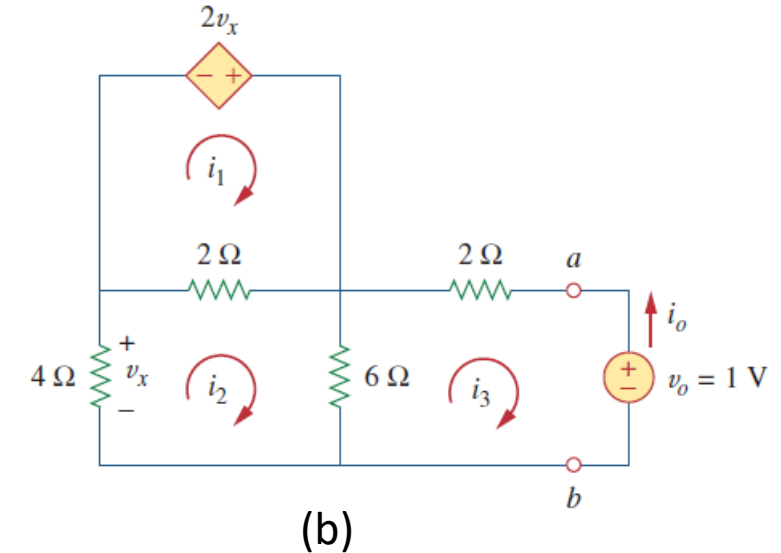
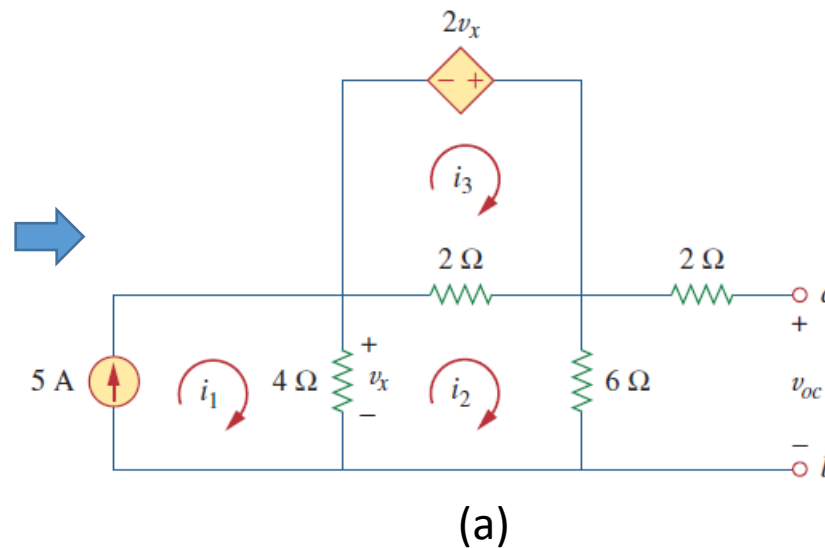


Figure 4.31
For Example 4.9.



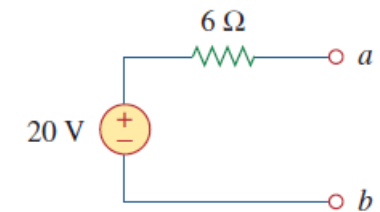
Solution:

Using the circuits shown above to find V_{Th} and R_{Th} in the Thevenin equivalent circuit

To find V_{Th} , mesh analysis for Fig. (a) finds $i_2 = 10/3$ A, so $V_{Th} = V_{oc} = 6i_2 = 20$ V

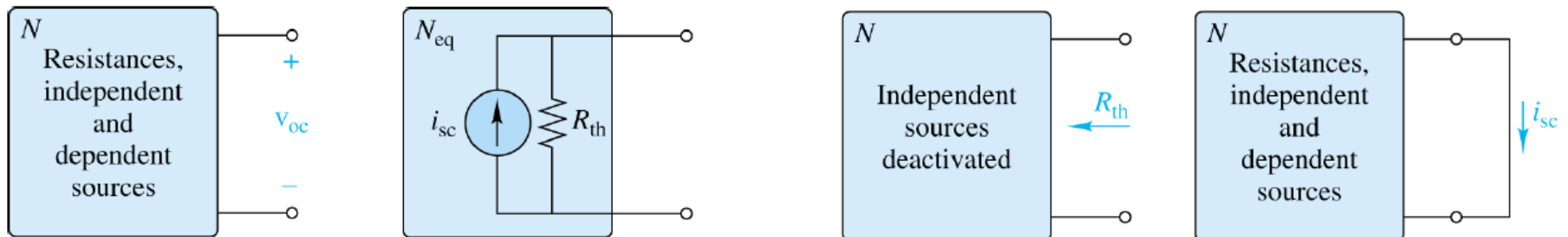
To find R_{Th} , apply a 1 V source at a-b in Fig. (b), we can find $i_o = 1/6$ A through mesh analysis, so $R_{Th} = 6$ Ω

Note that the VCVS were intact during both two processes



Norton's Theorem (I)

- In 1926 Edward Lawry Norton (1898-1983) a Bell Labs engineer came up with a similar theorem (think of source transformation)
- Norton's theorem (诺顿定理)** states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N
 - I_N is the **short-circuit current (短路电流)** through the terminals
 - R_N is the **input or equivalent resistance** at the terminals when the **independent sources are turned off**
- From source transformation, the Norton resistance will be exactly the same as the Thevenin, i.e., $R_N = R_{Th}$



Norton's Theorem (II)

- Norton's and Thevenin's theorems are closely related to each other
- One need only look at source transformation to understand this: $R_N = R_{Th}$, and $I_N = V_{Th}/R_{Th}$. For this reason, **source transformation (电源变换)** is often called **Thevenin-Norton transformation (戴维南-诺顿变换)**

$$R_N = R_{Th}$$

$$I_N = \frac{V_{Th}}{R_{Th}}$$

- With V_{Th} , I_N , and ($R_{Th} = R_N$) related, finding the Thevenin or Norton equivalent circuit requires that we find
 - The **open-circuit voltage** v_{oc} across terminals a and b
 - The **short-circuit current** i_{sc} at terminals a and b
 - The **equivalent or input resistance** R_{in} at terminals a and b
- The open-circuit and short-circuit tests are sufficient to find any Thevenin or Norton equivalent, of a circuit which contains at least one independent source

$$\begin{aligned} V_{Th} &= v_{oc} \\ I_N &= i_{sc} \\ R_{Th} &= \frac{v_{oc}}{i_{sc}} = R_N \end{aligned}$$

Example: Norton's Theorem (I)

- Find the Norton equivalent circuit of the circuit in Fig. 4.39 at terminals a-b

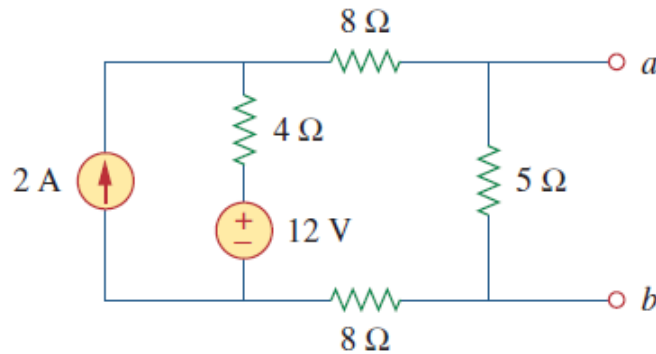


Figure 4.39

Solution:

To find R_N , turn off the independent sources, i.e., open-circuit the 2 A current source and short-circuit the 12 V voltage source, then $R_N = 5 \parallel (8+4+8) = 4 \Omega$

To find I_N , short the terminal a to b. Use superposition principle to find I_N as $I_N = 2 \cdot 4 / (4+8+8) + 12 / (4+8+8) = 1 \text{ A}$

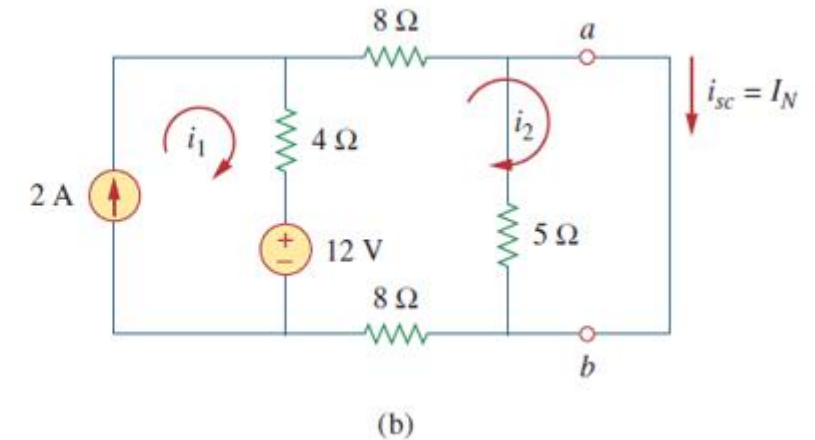
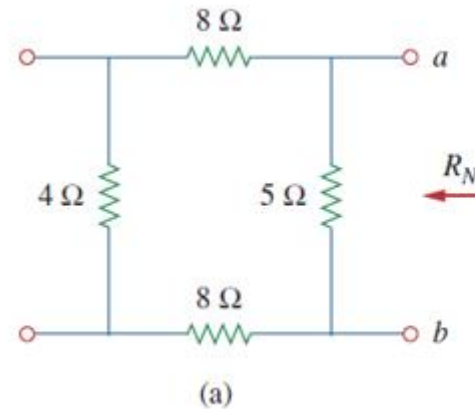


Figure 4.40

For Example 4.11; finding: (a) R_N , (b) $I_N = i_{sc}$

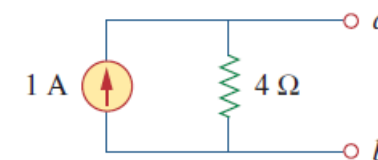


Figure 4.41

Norton equivalent of the circuit in Fig. 4.39.

Example: Norton's Theorem (II)

- Using Norton's theorem, find R_N and I_N of the circuit in Fig. 4.43 at terminals a-b

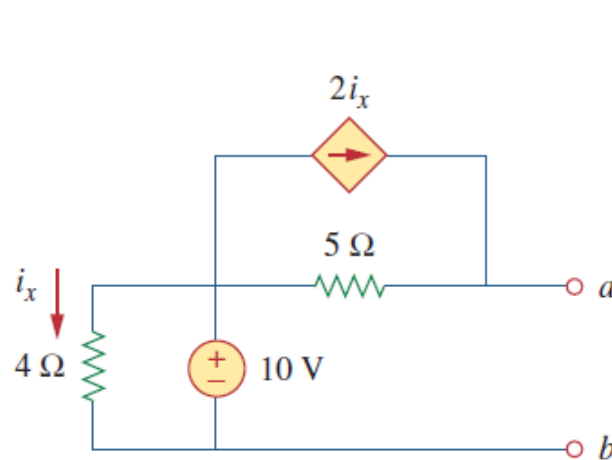


Figure 4.43

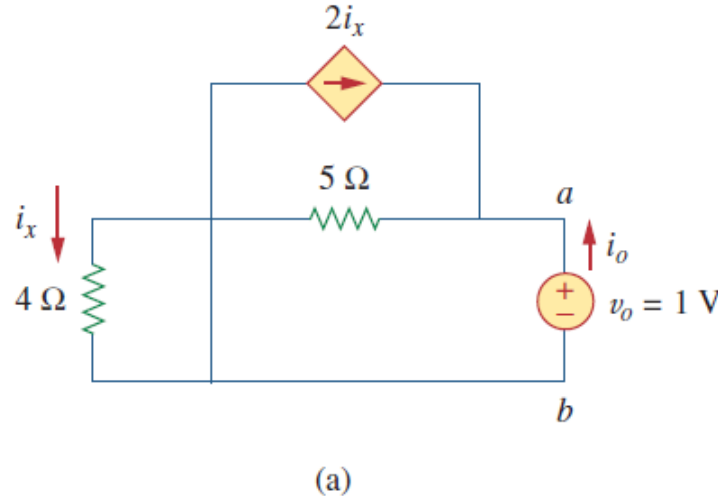
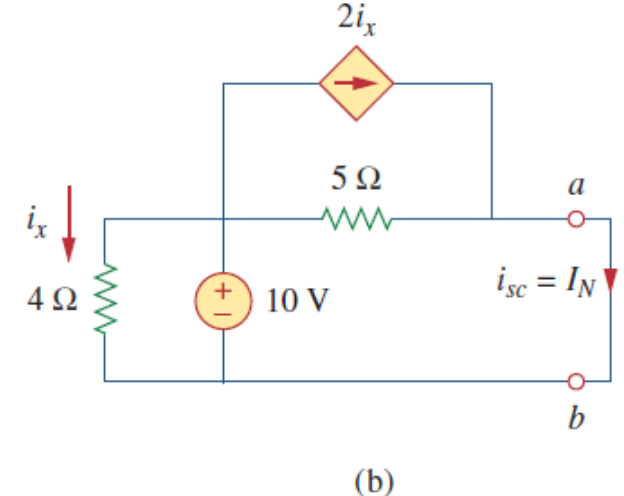


Figure 4.44

For Example 4.12: (a) finding R_N , (b) finding I_N .



Solution:

To find R_N , turn off the independent source, i.e., short-circuit the 10 V voltage source. Since the 4 Ω resistor is now short-circuited, $i_x = 0$ A, so does the CCCS. Therefore, $R_N = 5 \Omega$

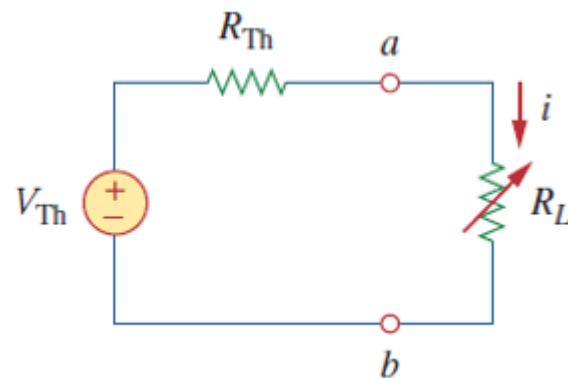
To find I_N , short the terminal a to b. Then $i_x = 10\text{V}/4\Omega = 2.5$ A. Therefore, $I_N = 2*i_x + 10\text{V}/5\Omega = 7$ A

Outline

- Linearity
- Superposition
- Source Transformation
- Thevenin's Theorem
- Norton's Theorem
- **Maximum Power Transfer**

Maximum Power Transfer (I)

- In many practical circuits, it is desirable to provide maximum power to the load
- Unlike an ideal source, **internal resistance (内部电阻)** will restrict the conditions where the maximum power is transferred
- We can **use the Thevenin equivalent** circuit for finding the maximum power in a linear circuit
 - Assume that the load resistance can be varied
 - Looking at the equivalent circuit with load included, the power transferred is

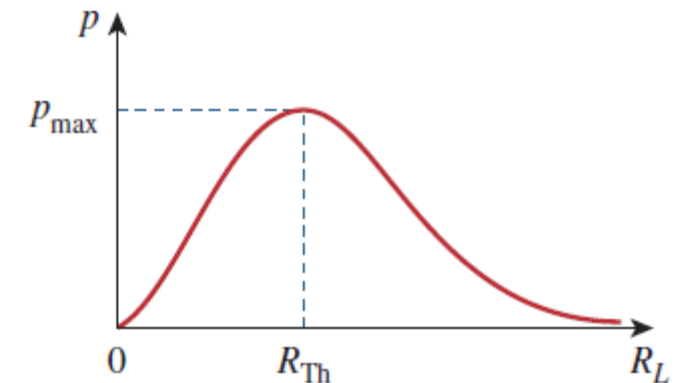


$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

Maximum Power Transfer (II)

- For a given circuit, V_{Th} and R_{Th} are fixed. By varying the load resistance R_L , the power delivered to the load varies
- Maximum power theorem (最大功率传输定理)** states that maximum power is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load (that is, $R_L = R_{Th}$)
 - To mathematically prove it, we differentiate power over R_L and set the result to zero

$$\begin{aligned} \frac{dp}{dR_L} &= V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right] \\ &= V_{Th}^2 \left[\frac{(R_{Th} + R_L - 2R_L)}{(R_{Th} + R_L)^3} \right] = 0 \end{aligned} \quad \Rightarrow \quad 0 = (R_{Th} + R_L - 2R_L) = (R_{Th} - R_L)$$



- Therefore, the maximum power transfer suggests that

$$R_L = R_{Th}$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

Example: Maximum Power Transfer

- Find the value of R_L for maximum power transfer in the circuit of Fig. 4.50. Find the maximum power

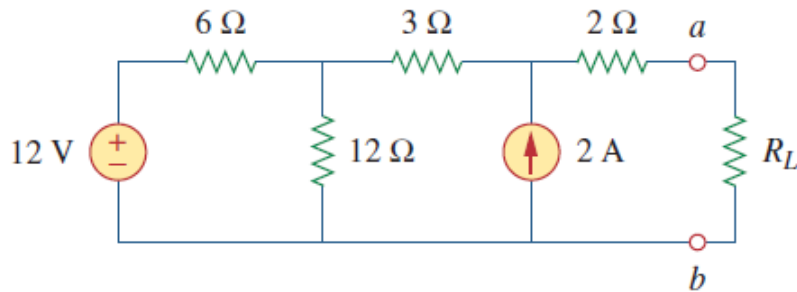


Figure 4.50

Solution:

We can find the Thevenin resistance R_{Th} and the Thevenin voltage V_{Th} across the terminals a-b

To find R_{Th} , turn off the independent sources, i.e., open-circuit the 2 A current source and short-circuit the 12 V voltage source. Then, $R_{Th} = 2 + 3 + 6 \parallel 12 = 9 \Omega$

To find V_{Th} , open-circuit the load R_L . Then according to superposition principle, we have $V_{Th} = 12 \cdot 12 / (12 + 6) + 2 \cdot (3 + 6 \parallel 12) = 22 \text{ V}$

Therefore, maximum power transfer requires that $R_L = R_{Th} = 9 \Omega$ and the corresponding p_{max} is

$$p_{max} = V_{Th}^2 / (4R_{Th}) = 13.444 \text{ W}$$

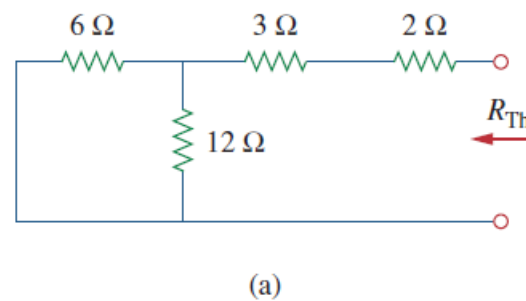
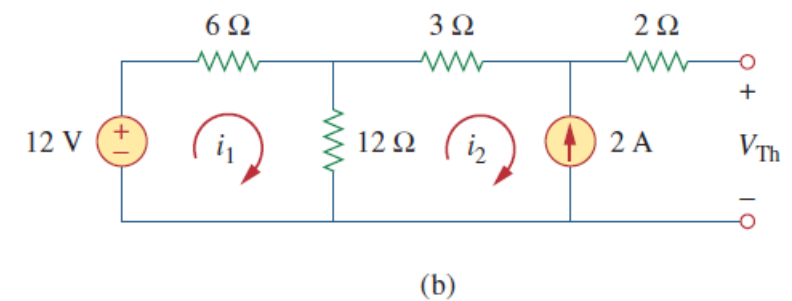


Figure 4.51

For Example 4.13: (a) finding R_{Th} , (b) finding V_{Th} .



Applications (Source Modeling)

- The Thevenin and Norton equivalent circuits are useful in understanding the behavior of realistic sources
 - An active source such as a battery is often characterized by its Thevenin or Norton equivalent circuit

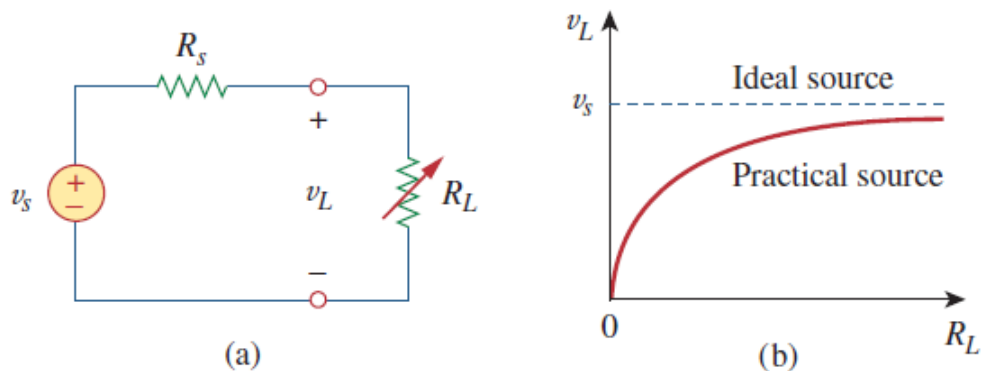


Figure 4.59

(a) Practical voltage source connected to a load R_L , (b) load voltage decreases as R_L decreases.

$$v_L = \frac{R_L}{R_s + R_L} v_s$$

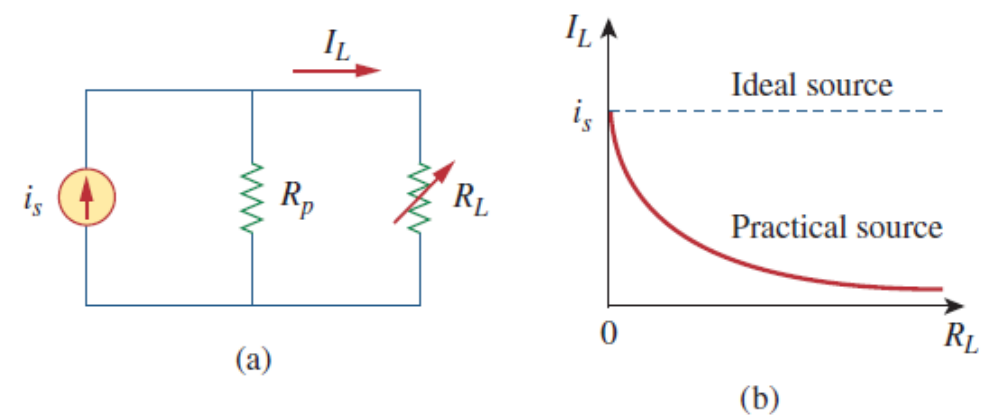


Figure 4.60

(a) Practical current source connected to a load R_L , (b) load current decreases as R_L increases.

$$i_L = \frac{R_p}{R_p + R_L} i_s$$

Summary

- A linear network consists of linear elements, linear dependent sources, and linear independent sources
- Network theorems are used to simplify a complex circuit, to make the circuit analysis easier
- The superposition principle states that for a circuit having multiple independent sources, the response (voltage across or current through) an element is equal to the algebraic sum of all the individual response due to each independent source acting one at a time
- Source transformation is a procedure for transforming a circuit consisting of a voltage source and a serial resistor to one with a current source and a parallel resistor, or vice versa
- Thevenin's and Norton's theorems allow us to isolate a portion of a network while the remaining portion of the network is replaced by an equivalent network. The Thevenin equivalent is in form of a voltage source V_{Th} in series with a resistor R_{Th} while the Norton equivalent consists a current source I_N and a parallel resistor R_N . The two theorems are related by the source transformation theorem
- For a given Thevenin equivalent circuit, maximum power transfer occurs when the load resistance equals to the Thevenin resistance of the circuit seen by the load, i.e., $R_L = R_{Th}$
- *Acknowledgement: This Lecture Note is partially based on materials from Prof. Chenchang Zan and partially adapted from the notes of the book "Fundamentals of Electric Circuits" by C. K. Alexander and M. N. O. Sadiku*